

WEEK-1

OUT-PUT EXERCISE-6

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:E:\IntelliJ2026\IntelliJ IDEA 2026.1.2\lib\idea_rt.jar=6344
Linear Search
102 Clean Code Robert Martin

Binary Search
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Book Not Found

Process finished with exit code 0
```

Explanation:

Time Complexity Comparison

Algorithm	Best Case	Average Case	Worst Case
Linear Search	$O(1)$	$O(n)$	$O(n)$
Binary Search	$O(1)$	$O(\log n)$	$O(\log n)$

When to Use Each Algorithm

Use Linear Search When:

- The dataset is small.
- The data is unsorted.
- Insertions and deletions occur frequently.
- Simplicity is preferred.

Use Binary Search When:

- The dataset is large.
- The data is already sorted.
- Fast searching is required.
- The collection changes infrequently